of vertices in that particular row and column. In our example, the adjacency matrix has an entry of 1 if and only if two people are friends. For example, the cell intersecting row B and column C has an entry of 1 because Biju and Chitra are friends (please verify this from Figure 1).

Now, consider the first row in the matrix given by Figure 3. What information do we get from it? We can see that there is no edge from vertex A to itself (this does not mean that Amrita is not friends with herself; this is just the convention I used for this example), hence the first entry is 0. The second entry is 1 because Amrita and Biju are friends. The third entry is 0 because Amrita and Chitra are not friends. The fourth entry is 1 because Amrita and Dev are friends. The fifth entry is 0 because Amrita and Ekta are not friends. The sixth entry is 1 because Amrita and Farhan are friends. Do verify these statements by looking at Figure 1, as understanding the adjacency matrix is very critical in working with graphs.

There is yet another way to represent graphs called an incidence matrix, which we will explore later in this article when discussing unweighted graphs with labels for edges.

The graph shown in Figure 1 (*Graph_1*) is an example of an undirected simple graph. A simple graph doesn't contain parallel edges or loops. Parallel edges are two or more edges having the same endpoints. A loop is an edge that starts and ends at the same vertex. For example, if you

	A	B	C	D	E	F
A	0	1	0	1	0	1
B	1	0	1	0	1	0
C	0	1	0	1	1	0
D	1	0	1	0	1	0
E	0	1	1	1	0	1
F	1	0	0	0	1	0

Figure 3: The adjacency matrix of *Graph_1*

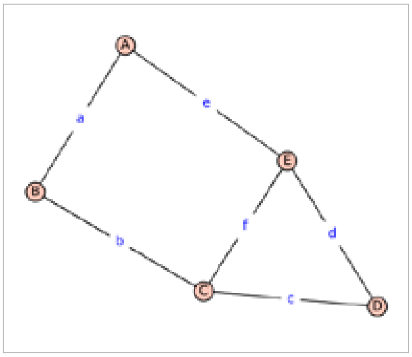

Figure 4: *Graph_2* generated by SageMath

want to explicitly show that Amrita is friends with herself, use an edge that starts and ends at vertex A. A graph is undirected if it only contains undirected edges. An undirected edge in a graph is like a two-way connection between two points. Think of our graph that represents friendships among people. If there's an undirected edge between two individuals, it means they are friends, and the friendship goes both ways. It is like saying if Amrita is friends with Dev, then at the same time, Dev is friends with Amrita. The connection is reciprocal, similar to a two-way friendship on social networking sites such as Facebook. It is important to distinguish this from platforms like Instagram, where you may follow a celebrity who may not reciprocally follow you back.

Now that we have a graph depicting the relationship status of six people, what possibilities does it open up for exploration or analysis? Well, once a real-world problem is modelled as a graph, a rich set of graph-theoretic tools can be employed to extract information from the graph. It is worth noting that graph theory is a very old branch of mathematics, and its origins can be traced back to the famous problem called the Seven Bridges of Königsberg and its resolution (in the negative) by the great mathematician Leonhard Euler in 1736.

Let us look at an example of using graph theory to solve a simple problem related to the six people. Suppose we need to identify the largest group of people among these six individuals who are friends with everyone else in this group. To find a solution to this problem, add the following lines of code at the end of *graph1.sagews* and execute it again.

```
friends = sgraph.cliques_maximum( )
print("The largest group of people who
are friends with each other: ", friends)
```

On execution, the following additional output will be printed on the screen. But what does it mean?

```
The largest group of people who are
friends with each other:  [['C', 'E',
'B'], ['D', 'C', 'E']]
```

The output indicates that Biju, Chitra, and Ekta form a group of three friends, and similarly, Chitra, Dev, and Ekta constitute another group of three friends. Moreover, it can be confirmed from the graph *Graph_1* that there is no group of size 4 where all four individuals are friends with each other. Solving a real-world problem was possible because we viewed it in terms of a graph-theoretic problem known as the clique problem. Formally, the clique problem is the computational problem of finding complete subgraphs in a graph. Please note that efficiently solving the clique problem is a well-known challenge in computer science. It falls into a category of problems referred to as NP-complete.

The power of a graph also derives from the fact that the same simple structure can easily abstract a lot of complicated real-world scenarios. For example, the vertices (A to F) in *Graph_1* could represent places in a city. Furthermore, the edges could signify the presence of a road connecting two places, and the weight of an edge could indicate the length of the road in kilometres. With one swift stroke, the graph denoting the relationship between people transforms into the graph of a city street map. Indeed, this exemplifies the beauty and